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WPC POWER DRIVER BOARD – COST–OPTIMISED MODERN VERSION – functional replacement for Williams A–12697–3 (same outline, connectors, fuses).
DPAK MOSFET solenoid/lamp drivers, TPS552892 buck–boost regulators for +5V/+12V, 4–quadrant insulated triacs, SMD passives, 5x20 mm fuses.
Sheets: Power_Supply, CPU_Interface, Sol_HighPower (1–8), Sol_LowPower (9–16), Sol_Flashers (17–24), Sol_GeneralPurpose (25–28),
Lamp_Columns, Lamp_Rows, GI_Triacs. Every part carries Manufacturer / MPN / DigiKey link / Price / Price100 fields (see BOM). See README.md.

Power_Supply

File: power_supply.kicad_sch

CPU_Interface

File: cpu_interface.kicad_sch

Sol_HighPower

File: sol_highpower.kicad_sch

Sol_LowPower

File: sol_lowpower.kicad_sch

Sol_Flashers

File: sol_flashers.kicad_sch

Sol_GeneralPurpose

File: sol_gp.kicad_sch

Lamp_Columns

File: lamp_columns.kicad_sch

Lamp_Rows

File: lamp_rows.kicad_sch

GI_Triacs

File: gi_triacs.kicad_sch

Modern re–implementation of the Williams A–12697–3 (same form factor)

Sheet: /
File: wpc_power_driver_cost.kicad_sch

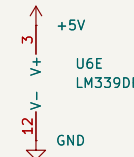
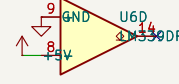
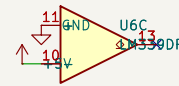
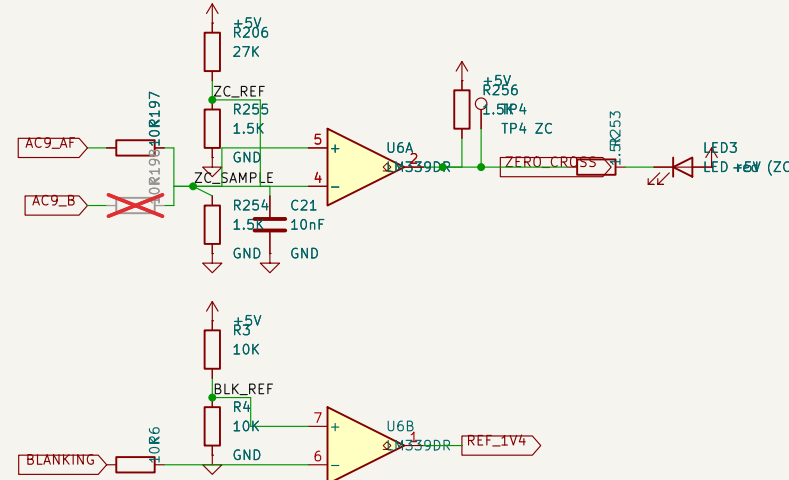
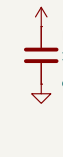
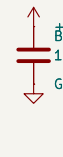
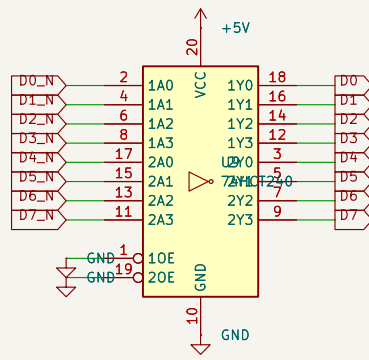
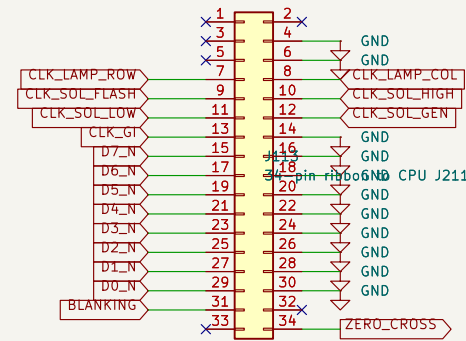
Title: WPC Power Driver Board – cost–optimised modern re–implementation

Size: A3Date: 2026–09–11Rev: JLC–3

KiCad E.D.A. 10.0.6Id: 1/10

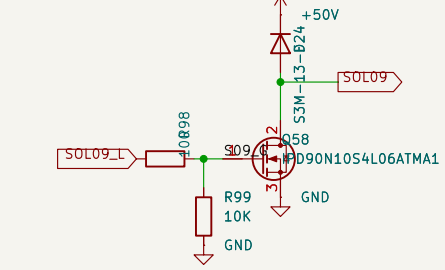
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CPU interface, pin-out from the WPC schematic manual 16-9834.2. J113: 7 /LMP ROW, 8 /LMP COL, 9 /SOL 2 (sol 17-24), 10 /SOL 4 (sol 1-8), 11 /SOL 3 (sol 9-16), 12 /SOL 1 (sol 25-28), 13 /TRIAC, 15,29 odd = /D7../D0, 31 BLANKING (HIGH = every latch tri-stated, open-collector from the CPU, pulled up here).

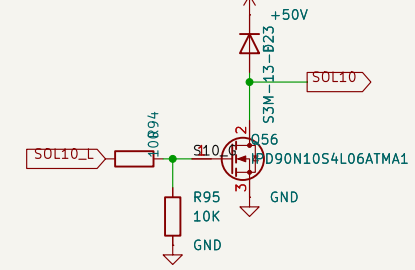


Low power solenoid drivers (sol. 9–16): IPD90N10S4L–06 logic-level DPAK MOSFETs (14 mOhm), latch U4 output HIGH = ON, clocked by /SOL 3 (J113–11). Tie-back diodes S3M (SMC, 3 A) to +50V (~70 V DC in the machine): coils in this group can be PWM-held by the firmware, the diode then carries the freewheel current. Output connectors J127 (playfield, key pin 2), J129 (backbox, sol 9–12, key 3) and J128 (cabinet, sol 13–16, key 4) as on the A–12697. Supply +50V via F104 on J107–2.

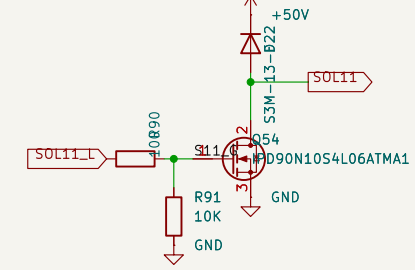
Sol 9 (low power)



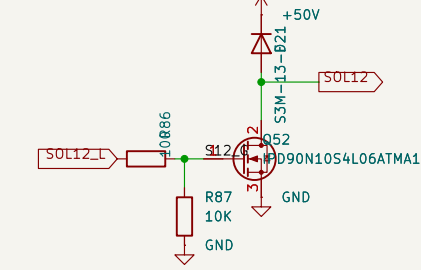
Sol 10 (low power)



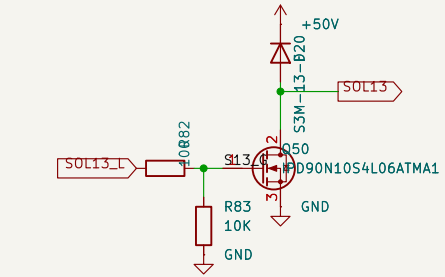
Sol 11 (low power)



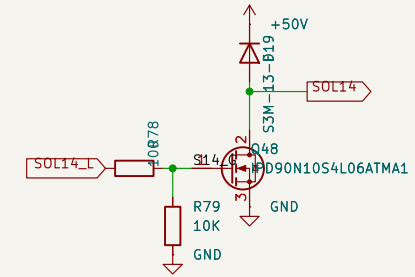
Sol 12 (low power)



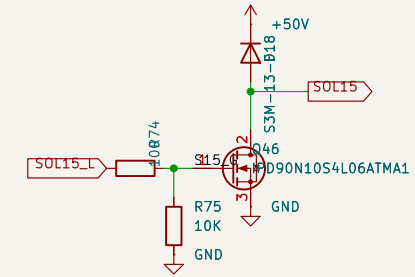
Sol 13 (low power)



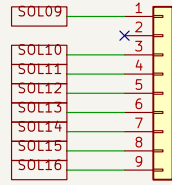
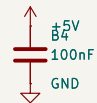
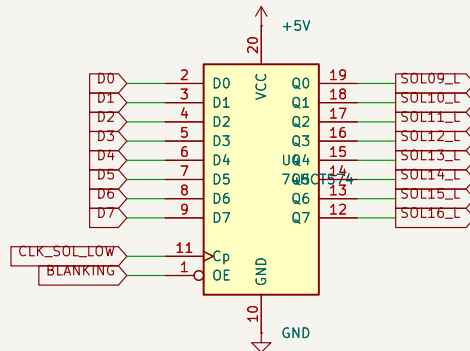
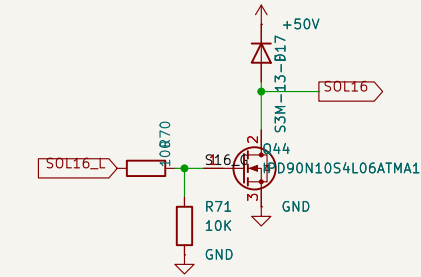
Sol 14 (low power)



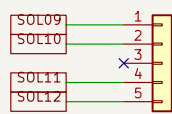
Sol 15 (low power)



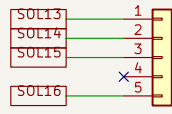
Sol 16 (low power)



J127 Sol 9–16 drive (playfield)



J129 Sol 9–12 drive (backbox, parallel to J127; N/C on STNG)



J128 Sol 13–16 drive (cabinet, parallel to J127; N/C on STNG)

Modern re-implementation of the Williams A–12697–3 (same form factor)

Sheet: /sol/LowPower/
File: sol_lowpower.kicad_sch

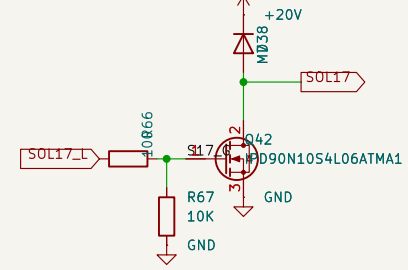
Title: Solenoids 9–16 (low power, IPD90N10S4L–06)

Size: A1 Date: 2026–09–11
Kicad E.D.A. 10.0.6

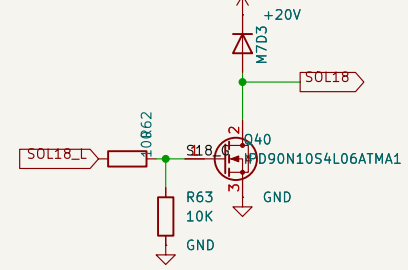
Rev: JLC–3
10/ 5/10

Flashlamp drivers (sol. 17–24): IPD90N10S4L–06 DPAK, latch U3 output HIGH = lamp ON, clocked by /SOL 2 (J113–9). Sol 17–20 keep an on-board S1M tie-back to +20V (the A–12697 has none there); the sol 21–24 tie-back cathodes are brought out to J126–10..13 exactly like the A–12697 D9–D12, so the harness chooses the clamp rail. On STING sol. 17/18 drive the 12 V gun motors (motor high side on the +12V power rail of J118) through this group. Outputs on J126 (playfield, key pin 9) and J125 (backbox, key pin 4). Flasher supply +20V on J107–5/6 / J106–5.

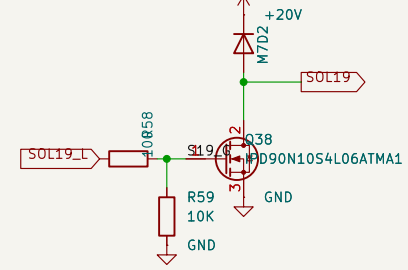
Sol 17 (flasher)



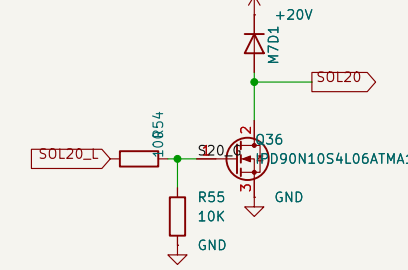
Sol 18 (flasher)



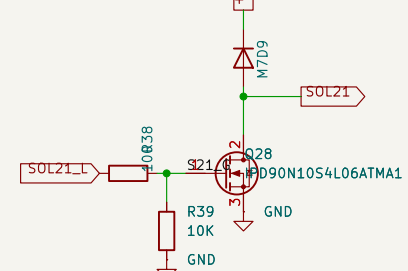
Sol 19 (flasher)



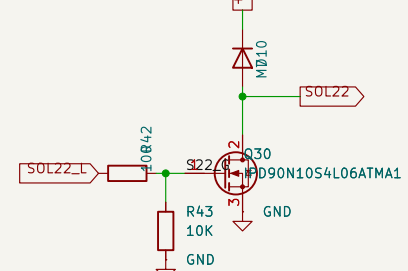
Sol 20 (flasher)



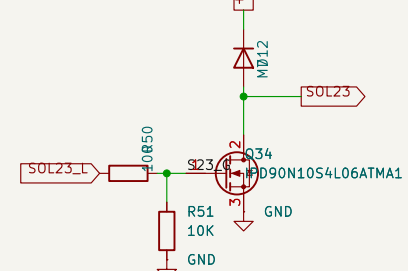
Sol 21 (flasher)



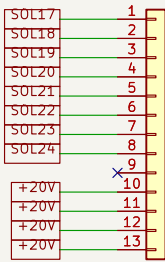
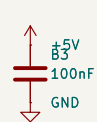
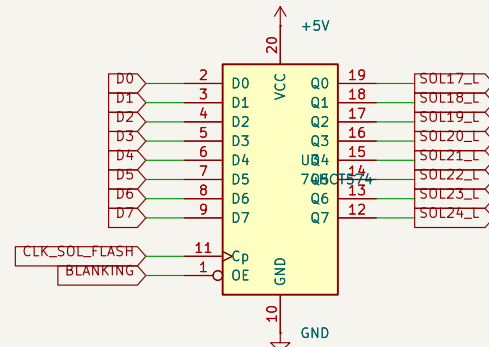
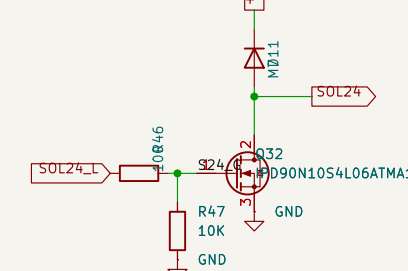
Sol 22 (flasher)



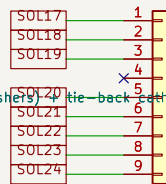
Sol 23 (flasher)



Sol 24 (flasher)



J126 Sol 17–24 drive (playfield flasher)



J125 Sol 17–24 drive (backbox flasher)

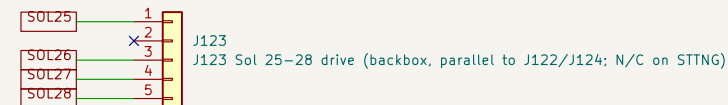
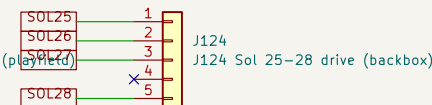
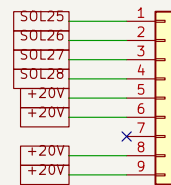
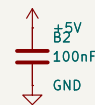
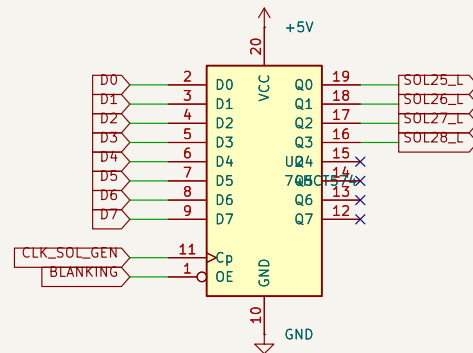
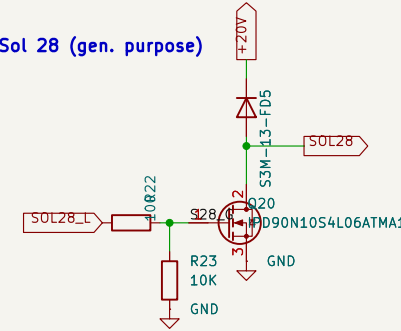
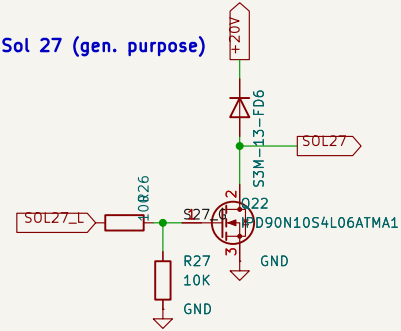
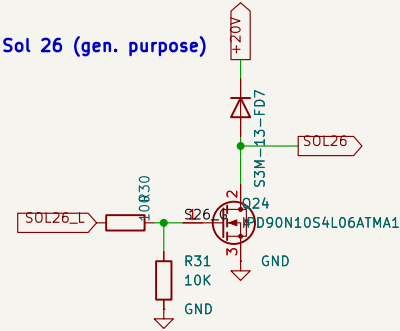
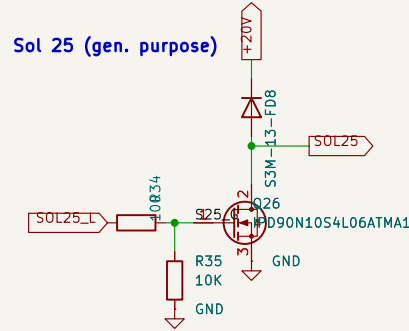
Modern re-implementation of the Williams A–12697–3 (same form factor)

Sheet: /sol/flashers/
File: sol_flashers.kicad_sch

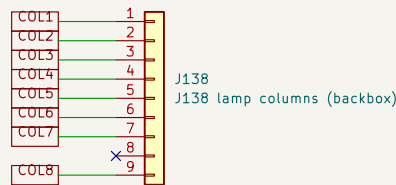
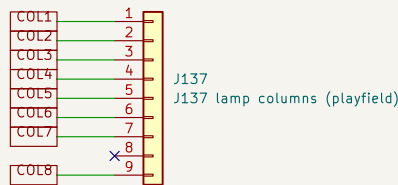
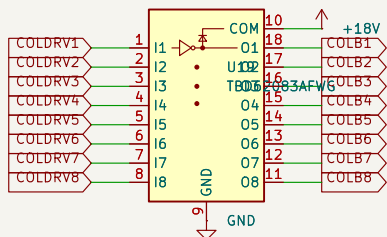
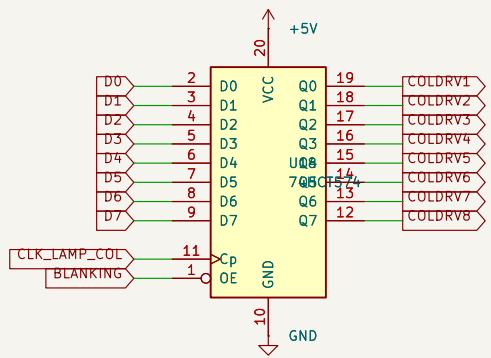
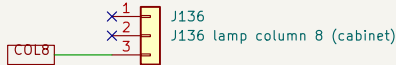
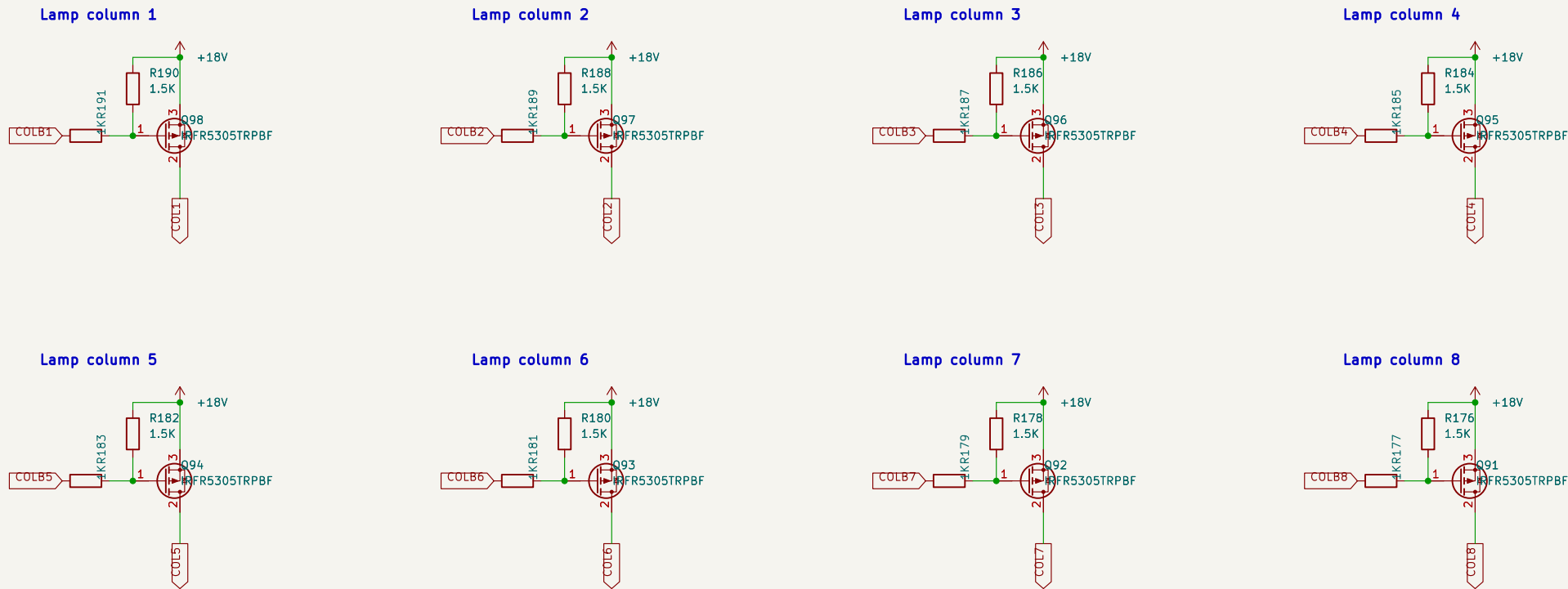
Title: Solenoids 17–24 (flashlamps, IPD90N10S4L–06)

Size: A1 Date: 2026–09–11 Rev: JLC–3
Kicad E.D.A. 10.0.6 Id: 6/10

General purpose drivers (sol. 25-28): IPD90N1054L-06 DPAK, latch U2 output HIGH = ON, clocked by /SOL 1 (J113-12). S3M tie-back diode cathodes are brought out to J122 pins 5,6,8,9 so the harness can tie them to the supply actually used (+50V coil or +20V flasher). Outputs J122 (playfield, key pin 7), J123 (backbox, key pin 2) and J124 (cabinet, key pin 4) as on the A-12697. U2 bits 4-7 unused.



Lamp matrix column drivers. Column strobe register 0x3FE5: bit(k-1)=1 selects column k, clocked by /LMP COL (J113-8). The TBD62083A inputs have an internal pull-down (Toshiba usage note), so a tri-stated U18 (BLANKING) leaves every column OFF – the OEM 74LS374/ULN2803 pair behaves the same way. U18 latches the data bus; the TBD62083A DMOS sink pulls the P-MOSFET gate down through 1 k against the 1.5 k gate-source resistor (Vgs = -0.6 x V+18 = -8..-12 V) -> column high (+18V) with -65 mOhm (IRFR5305 DPAK; TIP107 dropped -1.2 V). Column k = Q(99-k): col1=Q98 ... col8=Q91. Outputs J137/J138 pins 1-7,9 (key pin 8).



Modern re-implementation of the Williams A-12697-3 (same form factor)

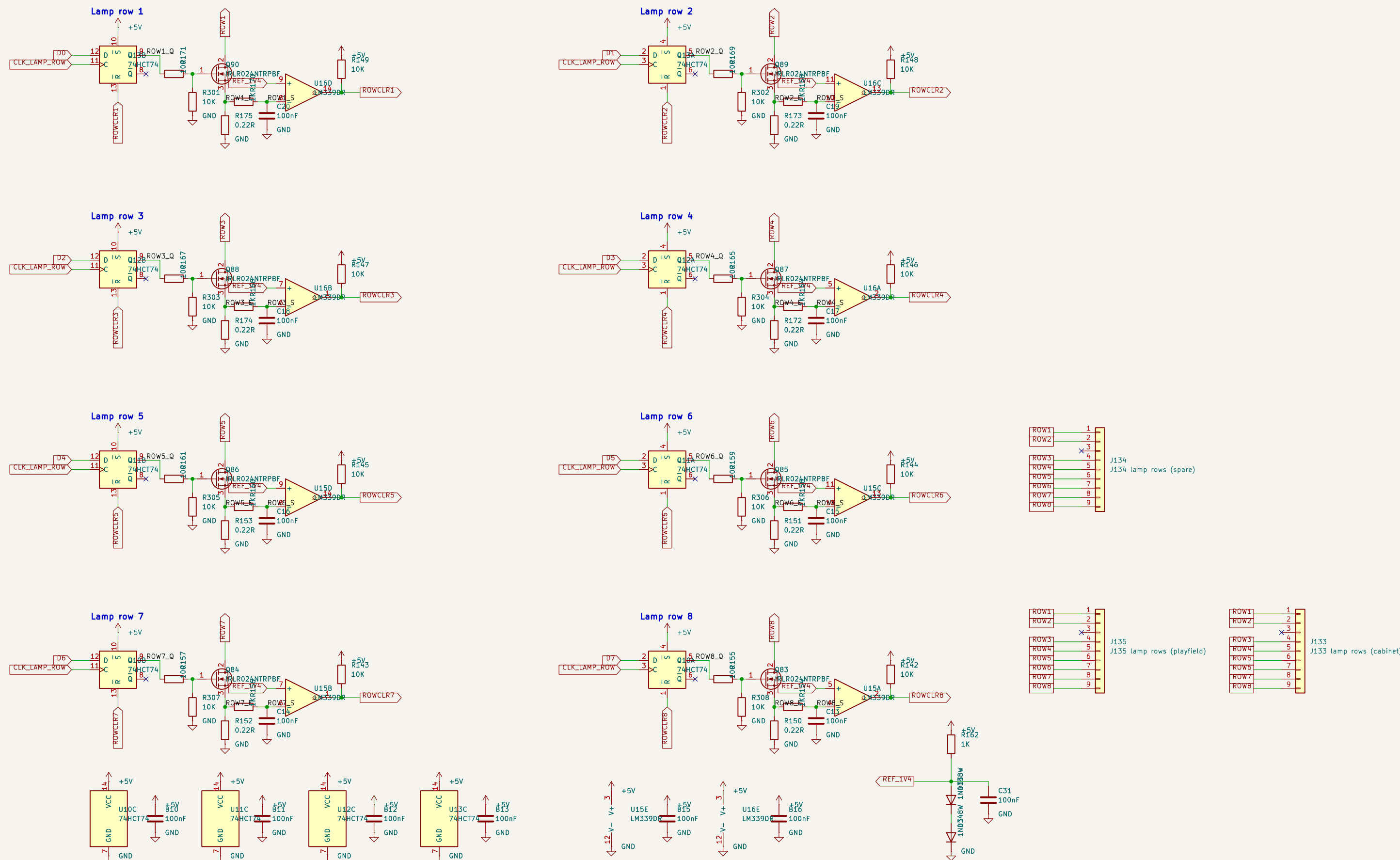
Sheet: /Lamp_Columns/
File: lamp_columns.kicad_sch

Title: Lamp matrix columns (IRFR5305 high side, +18V)

Size: A2 Date: 2026-09-11
KiCad E.D.A. 10.0.6

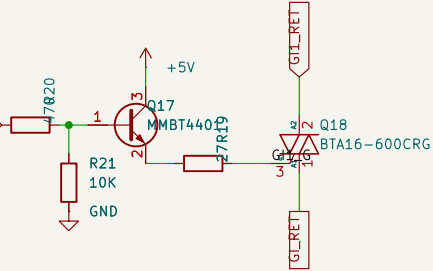
Rev: JLC-3
Id: 8/10

Ramp matrix row drivers. Row register 03FEF: bit(k-4) is turns row k on, clocked by /LMP_ROW_1J133-7. The 74HC714 latches the data bit (true after U9); Q high -> IRLR024N on (p = ground, -65-80 mVrms DPAK). Overcurrent: source sense 0.22 R -> 1k/100n -> LM339; above the 1.4 V reference (6 A) the open-collector outputs pins -CLR low and the row is shut off until the next strobe. BLANKING pins REF_1V4 low (CPU sheet) so all rows are forced off.
 Row k = 0: row1=Q90 ... row8=Q83. Outputs 1J33/1J35 (pins 1.2,4-9; key 3).

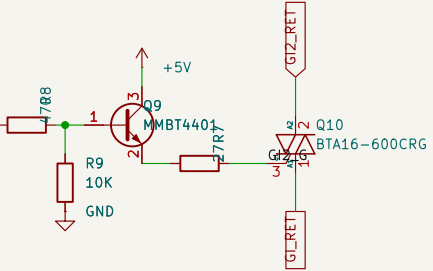


General illumination: each 6.3VAC string is fused (F106-F110) on the hot side and switched by a triac in the RETURN path; MT1 of the five triacs form the G_LRET bus that goes back to the transformer on J115-7/8/10/11/12 and is referenced to logic ground by W1 (OEM: J115-1 GND REF), so the 5 x 5 A return current stays off the ground plane. Latch U1 (clocked by TRIAC, J113-13) output HIGH = string ON. The gate is fed by an NPN emitter follower from +5V through 27 R (-80 mA, positive gate current in quadrants I and IV -> 4-quadrant "C" grade triac; snubberless BW/CW types cannot fire in Q IV). 10 k base pull-downs keep the strings OFF while the latch is tri-stated by BLANKING. Bits 5-7 (T5-T7) go to J111 (GPIO, CPU sheet); bit 7 also drives the optional flipper relay (DNP).

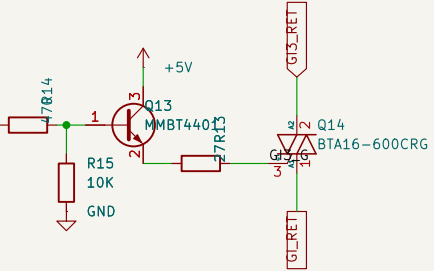
G.I. string #1 (Wht-Brn / Brn)



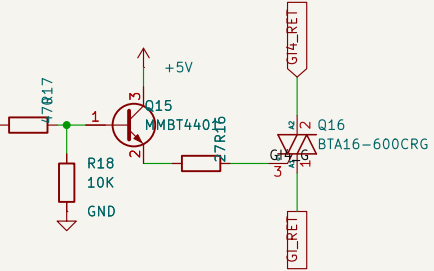
G.I. string #2 (Wht-Org / Org)



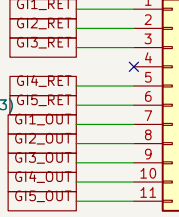
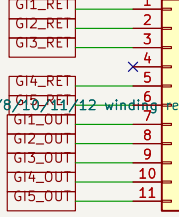
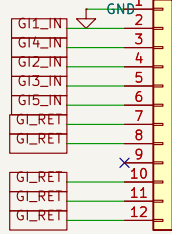
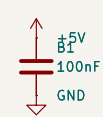
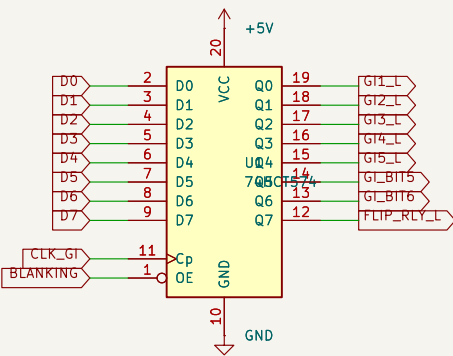
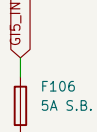
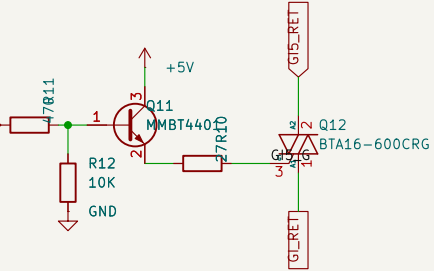
G.I. string #3 (Wht-Yel / Yel)



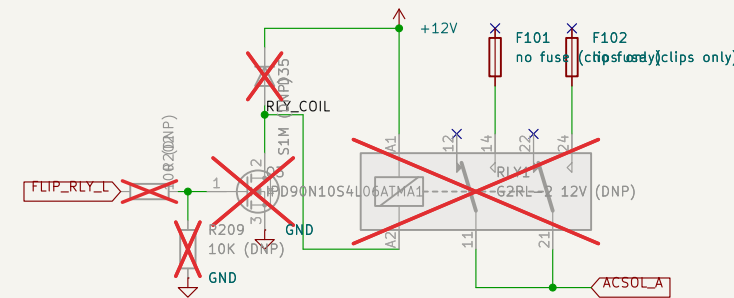
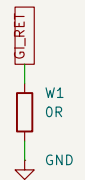
G.I. string #4 (Wht-Grn / Grn)



G.I. string #5 (Wht-Vio / Vio)



J121 J122 G.I. strings 1-5 (playfield harness uses strings 1,4,5)



Modern re-implementation of the Williams A-12697-3 (same form factor)

Sheet: /G.I./Triacs/

File: g.l.triacs.kicad.sch

Title: General illumination triacs + flipper relay option

Size: A1 Date: 2026-09-11

KiCad E.D.A. 10.0.6

Rev: JLC-3

10/10/25